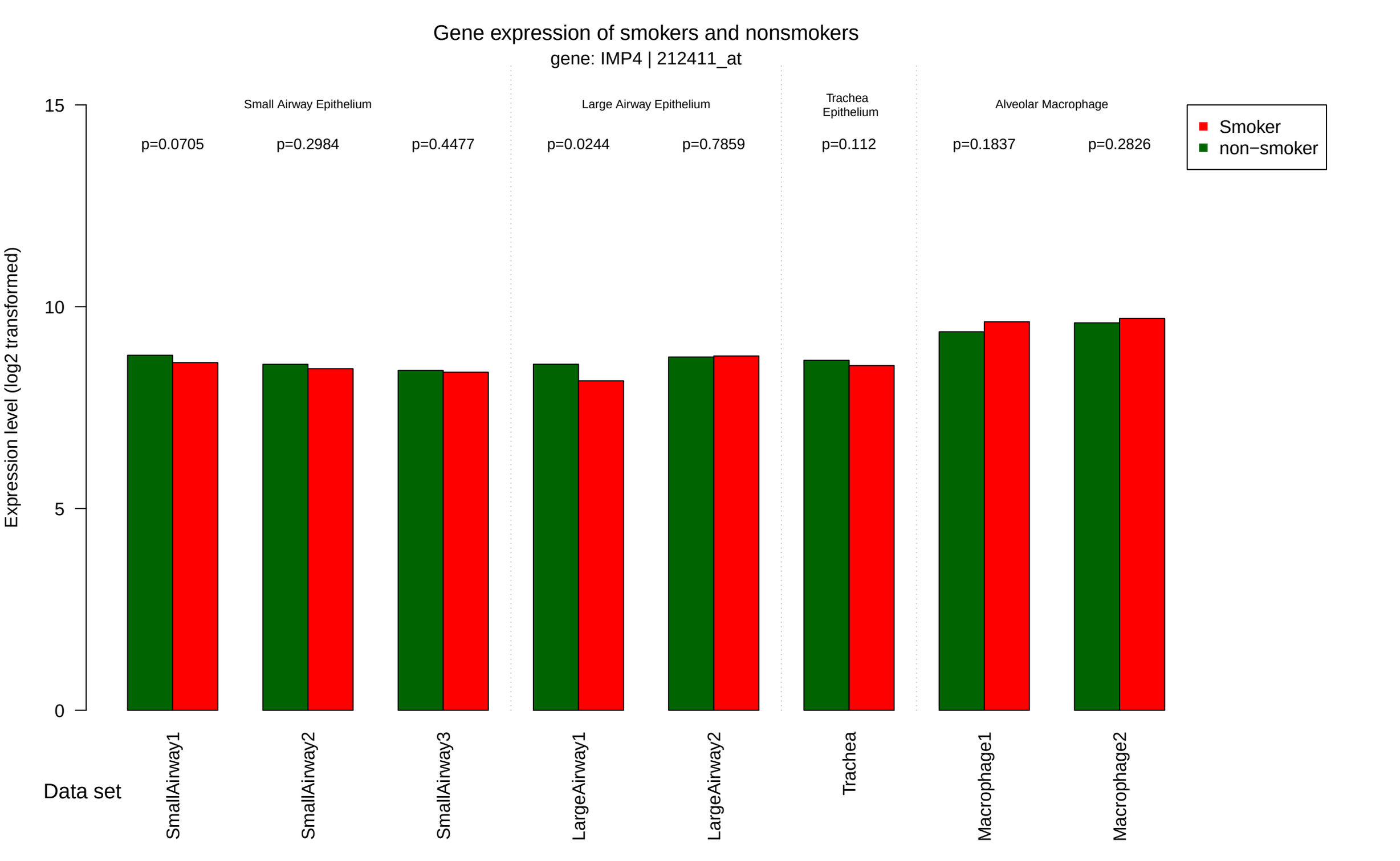
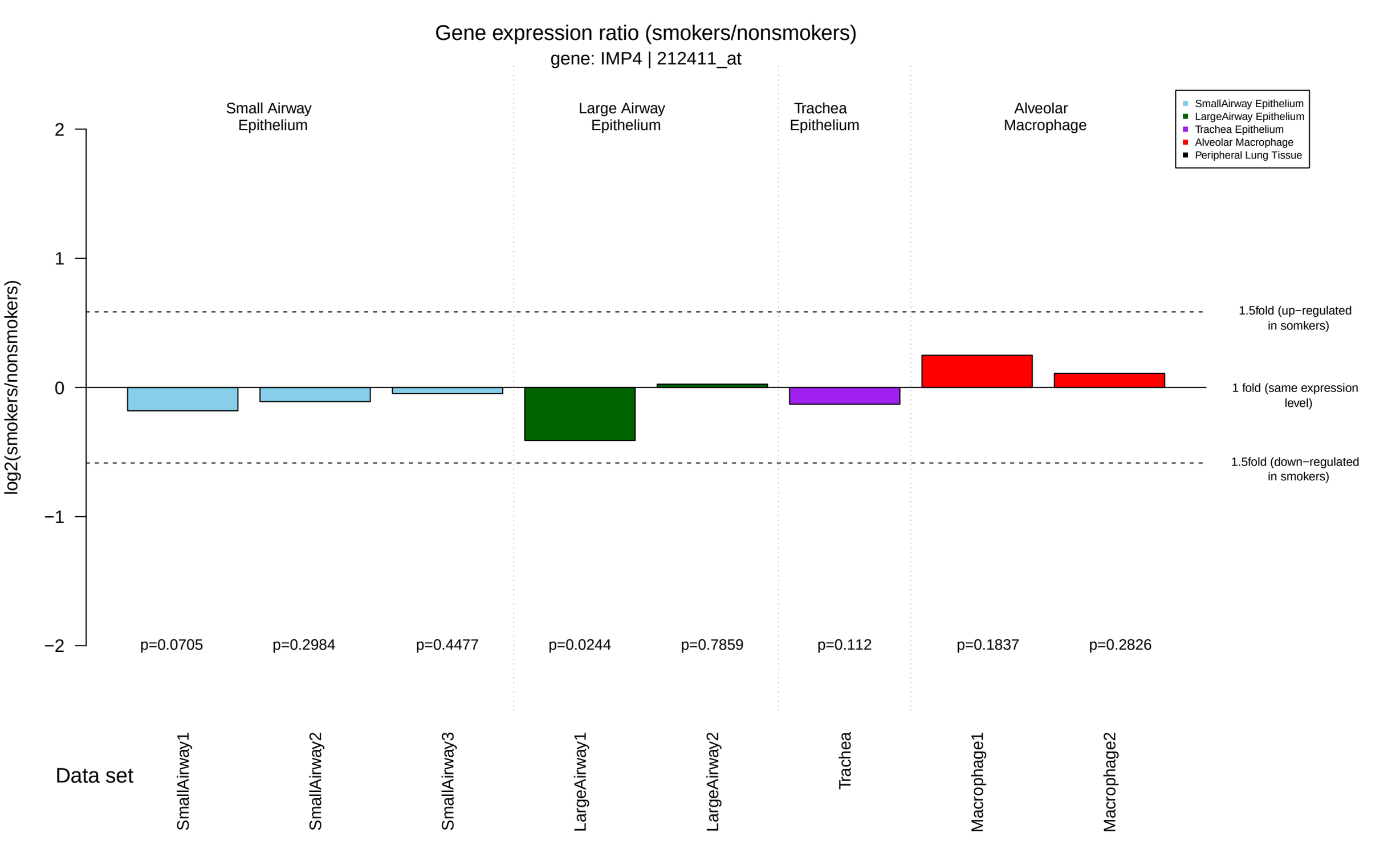
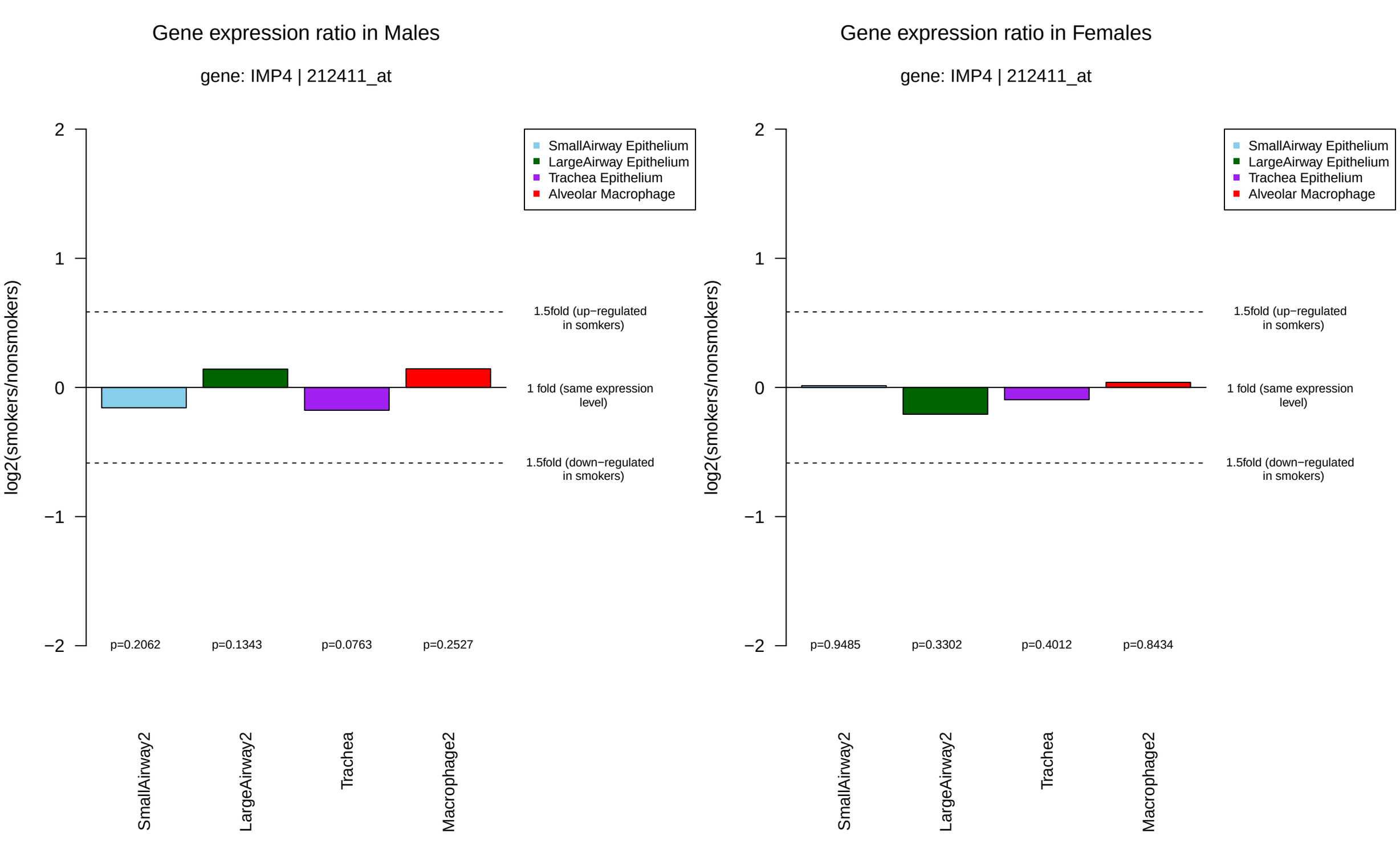
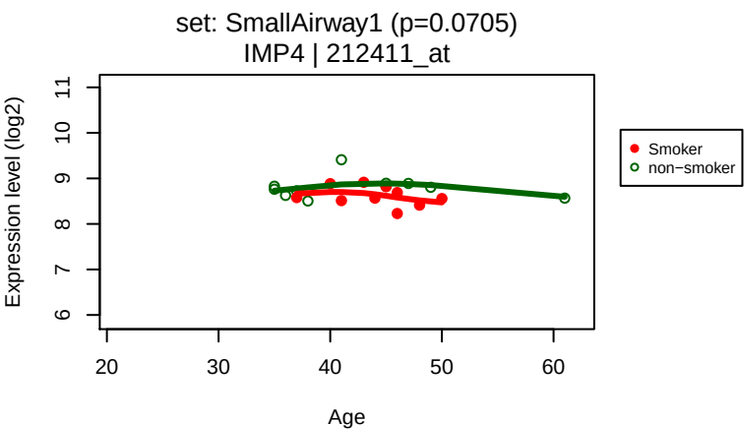
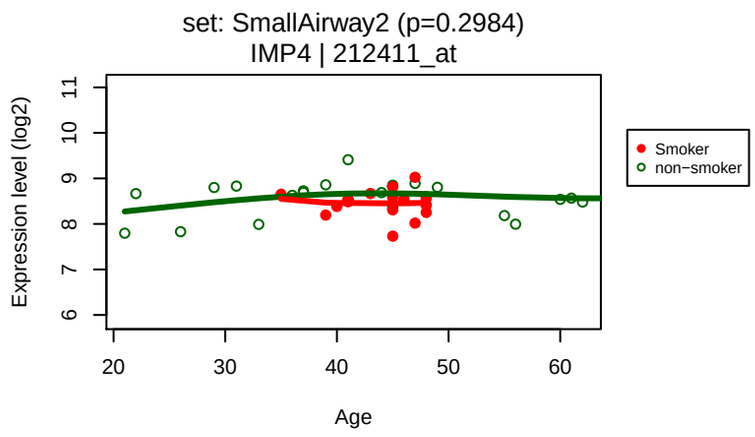
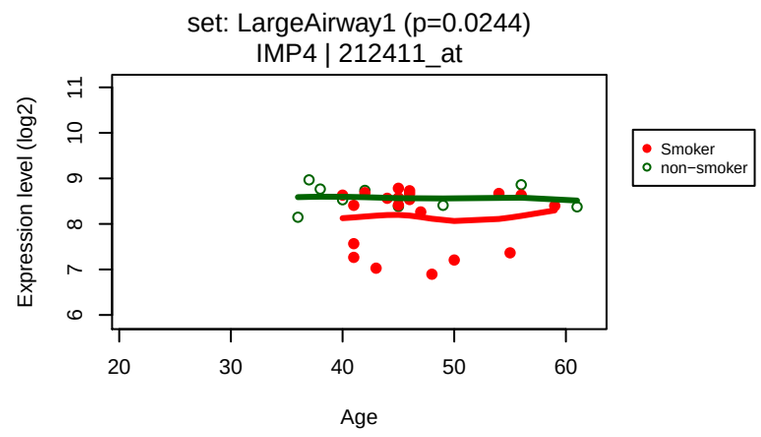

Gene expression of smokers and nonsmokers
gene: IMP4 | 212411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0705
p=0.2984
p=0.4477
p=0.0244
p=0.7859
p=0.112
p=0.1837
p=0.2826
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

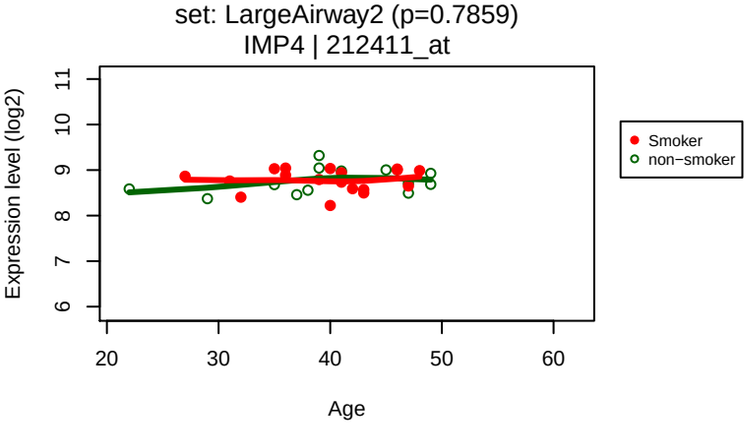

Gene expression ratio (smokers/nonsmokers)
gene: IMP4 | 212411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0705
p=0.2984
p=0.4477
p=0.0244
p=0.7859
p=0.112
p=0.1837
p=0.2826
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

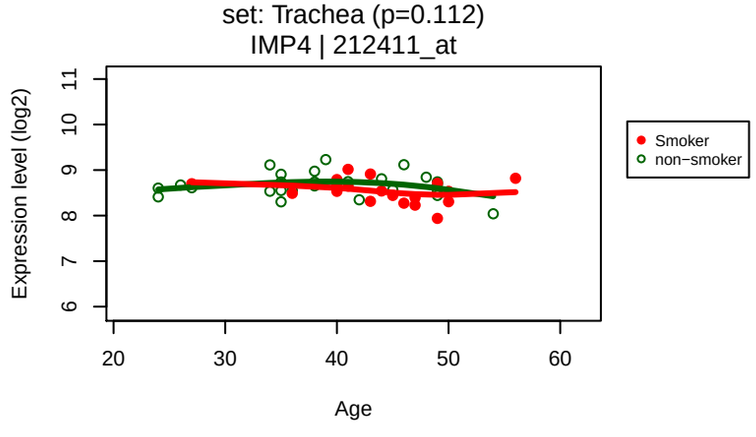

Gene expression ratio in Males
gene: IMP4 | 212411_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2062
p=0.1343
p=0.0763
p=0.2527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMP4 | 212411_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9485
p=0.3302
p=0.4012
p=0.8434
SmallAirway2
LargeAirway2
Trachea
Macrophage2

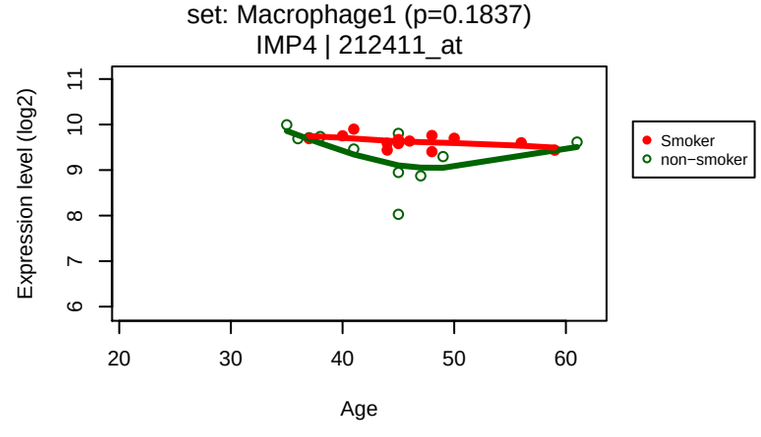

set: SmallAirway1 (p=0.0705)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker

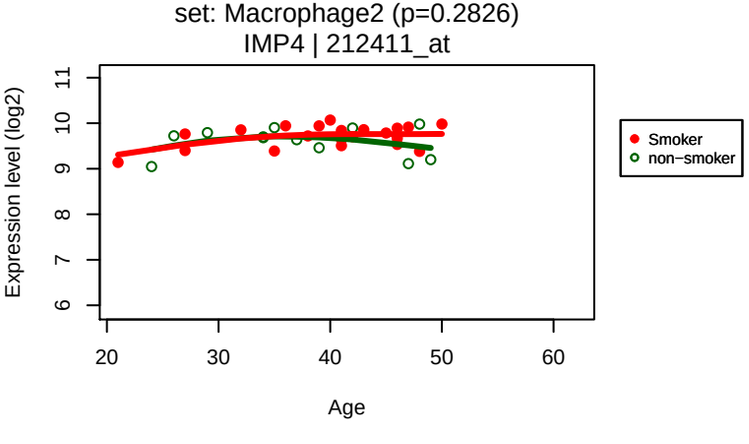

set: SmallAirway2 (p=0.2984)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker

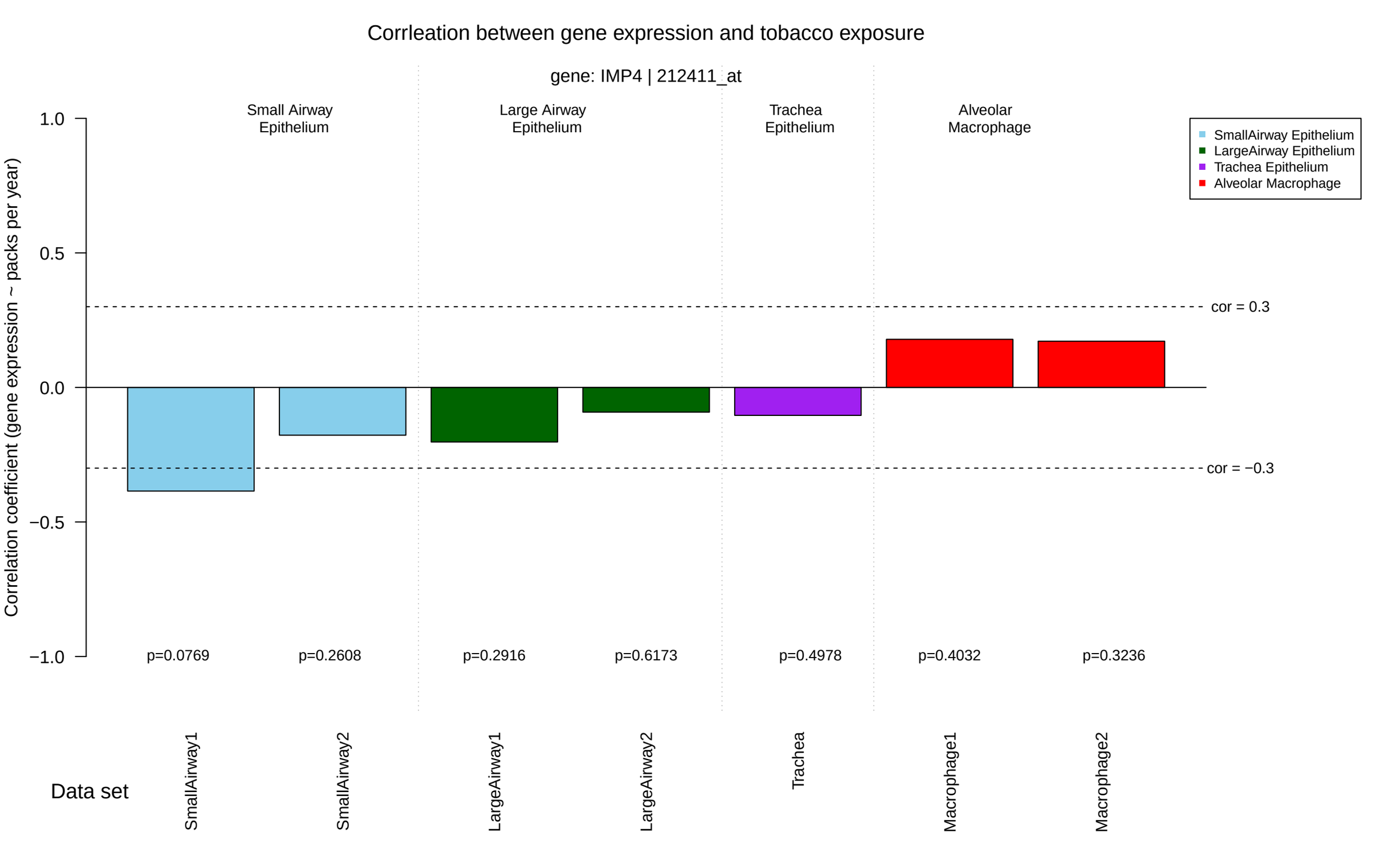

set: LargeAirway1 (p=0.0244)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7859)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.112)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1837)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2826)
IMP4 | 212411_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMP4 | 212411_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0769
p=0.2608
p=0.2916
p=0.6173
p=0.4978
p=0.4032
p=0.3236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2